MUD-UIE: Multimodal Unsupervised Depth-Guided Learning for Underwater Image Enhancement

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***Abstract*—Underwater images are severely degraded by light absorption, scattering, and depth-dependent color distortion, which significantly affects the performance of vision-based systems. Most underwater image enhancement (UIE) methods rely on supervised learning with paired reference images, a requirement that is rarely satisfied in real-world underwater environments. In this paper, we propose MUD-UIE (Multimodal Unsupervised Depth-Guided Learning for Underwater Image Enhancement), a fully unsupervised approach that enhances underwater images without ground-truth references. The proposed method jointly exploits RGB images and estimated depth maps within a multimodal learning strategy, enabling the model to learn scene-adaptive enhancement guided by no-reference image quality metrics. By incorporating depth-aware representations, MUD-UIE effectively captures scene-dependent degradation patterns and improves generalization across diverse underwater conditions. Experimental results on the UIEB, UVEB, and LSUI datasets show that MUD-UIE achieves the highest UIQM scores (5.501, 5.201, and 5.185) across all scenarios. Furthermore, it reaches superior PSNR and SSIM performance on all benchmarks while achieving competitive results in UCIQE and UISM, demonstrating the effectiveness of the proposed unsupervised depth-guided multimodal learning approach.**

***Index Terms*—Underwater Image Enhancement, Unsupervised Learning, Multimodal Learning, Depth-Guided Enhancement**

I. Introduction

Reliable visual perception in underwater environments is fundamental for applications such as autonomous navigation, marine inspection, and environmental monitoring, where visual sensors often serve as the primary source of information [1], [2]. The effectiveness of these high-level vision tasks strongly depends on image quality, making underwater image enhancement a critical component for robust perception rather than a simple preprocessing step [3].

Underwater images are degraded by wavelength-dependent light absorption and scattering effects, leading to color distortion, contrast loss, non-uniform illumination, and reduced visibility, particularly with increasing depth [4]. These degradations vary across water types and scenes, creating ambiguity among depth, illumination, and turbidity effects, which increases the difficulty of underwater image enhancement compared to terrestrial image restoration [5], [6].

Existing underwater image enhancement methods include physics-based models, prior-driven techniques, and learning-based approaches. While recent deep learning methods have shown promising results, supervised approaches rely on paired reference data that are rarely available in real underwater scenarios. Current unsupervised and no-reference methods, on the other hand, often suffer from perceptual inconsistencies, unstable enhancement behavior, and limited generalization across datasets [7], [8].

To address these challenges, we propose *MUD-UIE* (Multimodal Unsupervised Depth-Guided Learning for Underwater Image Enhancement), a fully unsupervised approach that integrates RGB images and estimated depth maps within a multimodal learning strategy. The proposed approach explicitly separates the definition of visual quality from the enhancement inference stage, allowing the model to learn enhancement behavior without relying on ground-truth references. Guided by blind-reference image quality metrics, MUD-UIE learns to predict enhancement strategies and their parameters in a continuous manner, enabling spatially adaptive and consistent enhancement across varying underwater conditions.

The main contributions of this paper are summarized as follows:

- A fully unsupervised underwater image enhancement approach guided by blind-reference quality assessment, eliminating the need for paired ground-truth images;
- A multimodal learning strategy that jointly exploits RGB appearance and depth information to predict both enhancement strategies and their parameters in a data-driven manner;
- A depth-guided mechanism that spatially modulates enhancement intensity, improving robustness and reducing perceptual artifacts in challenging underwater scenes.

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II. Related Work

Enhancing visual quality in underwater imagery remains a challenging task, due to the high variability of degradation patterns observed in different environments and acquisition conditions [9]. Early approaches relied on physics-based image formation models and handcrafted priors to compensate for color distortion and visibility loss [10]–[13]. While effective in controlled conditions, these methods often struggle to generalize across diverse underwater environments with varying water types and illumination conditions [6], [14], [15].

Most recent UIE methods adopt supervised learning with convolutional or transformer-based architectures, achieving strong performance through multi-scale attention, frequency modeling, or state-space representations [3], [16]–[18]. However, their dependence on paired or synthetic reference images limits practical applicability in real underwater scenarios. To alleviate this issue, semi-supervised and unsupervised strategies have been explored, including contrastive learning, consistency regularization [19], [20], and diffusion-based models [4], [5]. Despite promising results, many of these methods rely on implicit image synthesis and often suffer from spatial inconsistency, over-enhancement, or limited control over enhancement behavior.

Depth-aware and multimodal UIE has emerged as an important direction due to the strong correlation between scene depth and degradation severity. Recent works integrate depth cues via RGB-D priors or depth-guided attention mechanisms to improve stability across depth ranges [5], [8]. Nonetheless, these approaches mainly focus on direct image reconstruction and typically use no-reference quality metrics only for evaluation, restricting their ability to learn optimal enhancement behavior in fully reference-free settings [21], [22].

In contrast, decision-oriented and hybrid techniques suggest that explicitly learning enhancement strategy selection and parameterization can improve robustness and interpretability [1]. Building on this insight, our work formulates UIE as a multimodal decision and parameter estimation problem guided by blind-reference quality assessment. By jointly exploiting RGB appearance and depth information, the proposed approach enables adaptive and fully unsupervised enhancement across diverse underwater environments.

III. Methodology

A wide range of underwater visual conditions leads to scene-dependent degradations that challenge the use of fixed enhancement strategies. Consequently, relying on a single strategy with fixed parameters is often insufficient to address diverse underwater conditions. Instead of directly synthesizing enhanced images, the proposed *MUD-UIE* approach adopts a decision-oriented perspective, where underwater image enhancement is formulated as the estimation of an appropriate enhancement strategy and its corresponding parameters, guided by appearance and depth cues.

As illustrated in Fig. 1, the proposed approach processes RGB images and estimated depth maps in parallel within a multimodal learning framework. RGB information captures global color and contrast distortions, while depth provides a structural prior related to degradation severity and distance-dependent attenuation. By jointly reasoning over these complementary modalities, MUD-UIE enables adaptive enhancement that remains consistent across varying underwater environments, without relying on ground-truth reference images.

A. Blind-reference quality-driven supervision

In the absence of reference images, MUD-UIE relies on blind-reference image quality assessment to define a notion of perceptual optimality. Given an underwater image I and its estimated depth map D, multiple enhancement candidates are generated by applying different enhancement strategies $t \in \mathcal{T}$ with varying parameter settings $\boldsymbol{\theta} = [\Delta h,\ \alpha_{\text{rc}},\ \alpha_{\text{wb}},\ c]^\top$, corresponding to hue shift (HS), red-channel compensation (RC), white balance (WB), and the CLAHE clip limit. The strategy set $\mathcal{T}$ combines these operations in different orders (RC→WB and WB→RC, each optionally followed by HS or by CLAHE and HS) and also includes the UWE algorithm [7]. To account for distance-dependent degradation, each candidate is spatially modulated by the depth map,

$$\hat{I} = \mathbf{W} \odot \mathcal{P}_t(I;\boldsymbol{\theta}) + (\mathbf{1} - \mathbf{W}) \odot I, \qquad \mathbf{W} = \mathbf{1} - \beta D, \quad (1)$$

where $\mathcal{P}_t$ is the enhancement operator of strategy t, $\odot$ denotes pixel-wise multiplication, and $\beta = 0.8$. Each candidate image is evaluated using a set of complementary no-reference quality metrics, capturing underwater-specific color characteristics (UCIQE [23]), statistical naturalness (NIQE [24] and BRISQUE [25]), and perceptual artifacts (PIQE [26]). These metrics are used to identify, for each image–depth pair, the enhancement configuration that yields the most favorable perceptual quality: after min–max normalization across candidates, the best candidate according to the average score and the best according to a color- and naturalness-weighted score (0.4 UCIQE, 0.4 NIQE, 0.2 BRISQUE) are compared, and the one with the lowest PIQE is retained.

This process provides pseudo-supervision in the form of an enhancement strategy t^* and a parameter vector $\boldsymbol{\theta}^*$, while maintaining an unsupervised learning setting. No human annotations or paired reference images are required during the definition of high-quality strategies and parameter settings.

B. Multimodal depth-guided learning

The core of the MUD-UIE is a multimodal neural network architecture that predicts enhancement decisions from RGB appearance and depth information. Two modality-specific encoders operate in parallel: a Vision Transformer processes the RGB image to capture global color and texture distortions, while a convolutional encoder extracts spatial and geometric cues from the estimated depth map. The depth map D is estimated from I using Depth Anything [27], a pre-trained monocular depth estimator that is kept frozen, and normalized to $[0, 1]$. These encoders produce the feature representations:

$$\mathbf{F}_{\text{rgb}} = \mathcal{E}_{\text{rgb}}(I), \qquad \mathbf{F}_{\text{depth}} = \mathcal{E}_{\text{depth}}(D). \quad (2)$$

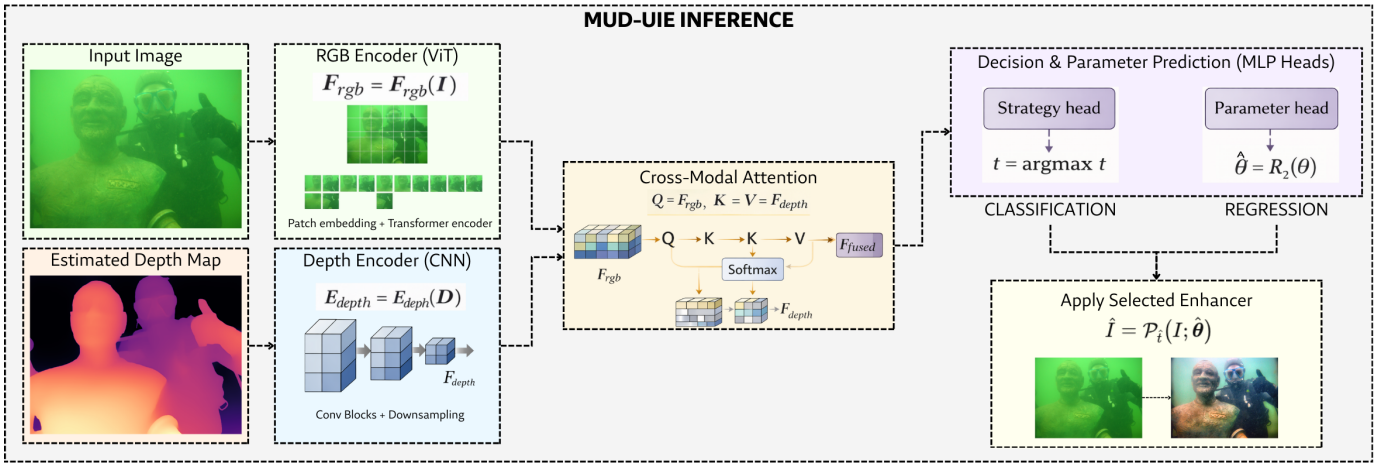


Fig. 1. Overview of the proposed MUD-UIE inference pipeline. RGB appearance and depth information are encoded in parallel and fused via depth-guided cross-modal attention to predict the enhancement strategy and its parameters for UIE.

To integrate appearance and geometry, MUD-UIE adopts a depth-guided cross-modal attention mechanism, in which RGB features act as queries and depth features provide keys and values. This formulation conditions enhancement decisions on depth-dependent degradation patterns:

$$\mathbf{F}_{\text{fused}} = \text{softmax}\left(\frac{QK^{\top}}{\sqrt{d}}\right)V, \quad Q = \mathbf{F}_{\text{rgb}},\ K = V = \mathbf{F}_{\text{depth}}. \tag{3}$$

From the fused representation $\mathbf{F}_{\text{fused}}$, the enhancement configuration is inferred through a multi-task decision module composed of multi-layer perceptron (MLP) heads. A classification head predicts the enhancement strategy,

$$\hat{t} = \arg\max_{t}\ p(t \mid \mathbf{F}_{\text{fused}}), \tag{4}$$

while a regression head estimates the enhancement parameters,

$$\hat{\boldsymbol{\theta}} = \mathcal{R}(\mathbf{F}_{\text{fused}}). \tag{5}$$

Both heads are trained jointly with the pseudo-labels $(t^*, \boldsymbol{\theta}^*)$ by minimizing

$$\mathcal{L} = \mathcal{L}_{\text{CE}}\big(p(t \mid \mathbf{F}_{\text{fused}}),\ t^*\big) + \lambda\, \tfrac{1}{4}\big\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*\big\|_2^2, \tag{6}$$

where $\mathcal{L}_{\text{CE}}$ denotes the cross-entropy loss and $\lambda = 1$.

In this design, the encoders provide modality-specific representations, while the MLP heads jointly perform enhancement strategy selection and parameter estimation. This formulation explicitly captures the coupling between which enhancement operation should be applied and how it should be parameterized for a given scene. At inference time, the predicted configuration $(\hat{t}, \hat{\boldsymbol{\theta}})$ is passed to the deterministic enhancement operator $\mathcal{P}_{\hat{t}}(I; \hat{\boldsymbol{\theta}})$ outside the neural network, and its output is spatially modulated by the depth map according to Eq. (1).

As a result, the final enhanced image is generated using the predicted optimal parameters, ensuring efficient inference while preserving the perceptual behavior learned through blind-reference supervision during training.

IV. Experiments

This section evaluates the proposed *MUD-UIE* approach through qualitative and quantitative experiments on well-established underwater image enhancement benchmarks under a fully no-reference protocol, without using paired ground-truth images during training or inference. The model was implemented in PyTorch and evaluated on a workstation equipped with an Intel® Xeon™ Silver 4114 CPU, 128 GB RAM, and an NVIDIA® RTX A4000 GPU (16 GB); the RGB branch adopts a ViT-Base backbone with 16×16 patches, depth is processed by a convolutional encoder, and both modalities are fused via cross-modal attention, followed by a multi-task head that jointly predicts the enhancement strategy and its parameters. Training was performed for 1000 epochs using AdamW with a learning rate of 5×10^{-5} and an 80/20 train–test split.

Experiments were carried out on the UIEB (Underwater Image Enhancement Benchmark), UVEB (Underwater Video Enhancement Benchmark), and LSUI (Large-Scale Underwater Image dataset) datasets, which cover diverse underwater conditions such as color casts, turbidity, low contrast, and depth-dependent degradation. The performance evaluation was conducted using recent supervised, semi-supervised, and unsupervised methods from top-tier venues, as reported in Tables I–III, using both full-reference metrics (PSNR and SSIM) and underwater-specific no-reference metrics, namely the Underwater Color Image Quality Evaluation (UCIQE) [23], the Underwater Image Quality Measure (UIQM) [29], and the Underwater Image Sharpness Measure (UISM) [29]. For UVEB, full-reference metrics were computed frame-wise against the reference frames provided with the benchmark and averaged over all test frames.

A. Quantitative Evaluation

Tables I–III present quantitative results on the UIEB, UVEB, and LSUI datasets using both full-reference (PSNR and SSIM) and underwater-specific no-reference metrics

TABLE I
QUANTITATIVE COMPARISON ON THE UIEB DATASET USING FULL-REFERENCE AND NO-REFERENCE EVALUATION METRICS.

Type	Method	Venue	PSNR↑	SSIM↑	UCIQE↑	UIQM↑	UISM↑
S	CE-VAE [2]	WACV'25	17.28	0.751	0.530	2.782	4.082
S	U-shape [16]	TIP'23	21.65	<u>0.962</u>	0.569	1.024	5.952
S	Spectroformer [3]	WACV'24	20.34	0.846	0.591	3.152	4.679
S	Phaseformer [17]	WACV'24	20.78	0.859	0.602	3.213	4.746
S	PixMamba [18]	ACCV'24	<u>22.87</u>	0.927	**0.782**	<u>5.088</u>	**6.561**
S	USIS10K [28]	ICML'24	20.56	0.852	0.598	3.185	4.712
SS	SEMI-UIR [19]	CVPR'23	20.69	0.952	0.719	4.647	5.968
U	TACL [20]	TIP'22	18.42	0.783	0.542	2.893	4.216
U	OSMOSIS [5]	ECCV'24	17.65	0.762	0.533	2.816	4.123
U	**MUD-UIE**	–	**22.94**	**0.981**	<u>0.732</u>	**5.501**	<u>6.513</u>

TABLE II
QUANTITATIVE COMPARISON ON THE UVEB DATASET UNDER DIVERSE UNDERWATER DEGRADATION CONDITIONS.

Type	Method	Venue	PSNR↑	SSIM↑	UCIQE↑	UIQM↑	UISM↑
S	CE-VAE [2]	WACV'25	17.45	0.760	0.541	2.803	4.115
S	U-shape [16]	TIP'23	21.84	**0.973**	0.579	1.057	5.993
S	Spectroformer [3]	WACV'24	20.51	0.853	0.603	3.179	4.723
S	Phaseformer [17]	WACV'24	20.92	0.868	0.616	3.240	4.792
S	PixMamba [18]	ACCV'24	<u>22.12</u>	<u>0.966</u>	**0.792**	<u>5.112</u>	**6.615**
S	USIS10K [28]	ICML'24	20.78	0.861	0.609	3.208	4.757
SS	SEMI-UIR [19]	CVPR'23	20.56	0.955	0.727	4.662	5.992
U	TACL [20]	TIP'22	18.67	0.792	0.553	2.916	4.268
U	OSMOSIS [5]	ECCV'24	17.52	0.763	0.539	2.823	4.146
U	**MUD-UIE**	–	**23.01**	**0.973**	<u>0.774</u>	**5.201**	<u>6.577</u>

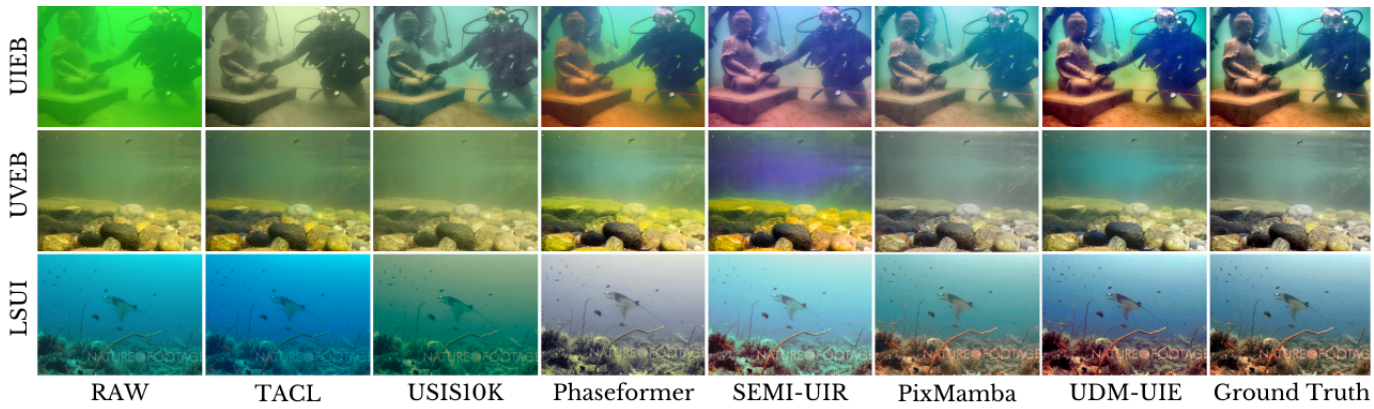


Fig. 2. Qualitative comparison on UIEB (first row), UVEB (second row), and LSUI (third row). From left to right: Input (RAW), TACL, USIS10K, Phaseformer, SEMI-UIR, PixMamba, MUD-UIE (ours), and Ground Truth.

(UCIQE, UIQM, and UISM). The comparison includes supervised (S), semi-supervised (SS), and unsupervised (U) methods, enabling a fair analysis across different supervision conditions. Across all benchmarks, MUD-UIE achieves superior full-reference performance while consistently maintaining strong no-reference scores. These results demonstrate that no-reference supervision combined with depth-guided multi-modal decision learning can deliver stable and high-quality enhancement without relying on paired reference images. Notably, MUD-UIE highlights its robustness and its strong

TABLE III
QUANTITATIVE COMPARISON ON THE LSUI DATASET, EMPHASIZING GENERALIZATION IN REAL-WORLD UNDERWATER SCENARIOS.

Type	Method	Venue	PSNR↑	SSIM↑	UCIQE↑	UIQM↑	UISM↑
S	CE-VAE [2]	WACV'25	17.12	0.745	0.527	2.765	4.057
S	U-shape [16]	TIP'23	21.48	**0.955**	0.562	1.003	5.923
S	Spectroformer [3]	WACV'24	20.18	0.839	0.587	3.131	4.653
S	Phaseformer [17]	WACV'24	20.61	0.851	0.599	3.192	4.723
S	PixMamba [18]	ACCV'24	<u>22.71</u>	0.919	**0.978**	<u>5.065</u>	<u>6.538</u>
S	USIS10K [28]	ICML'24	20.39	0.845	0.593	3.163	4.692
SS	SEMI-UIR [19]	CVPR'23	20.41	<u>0.946</u>	0.716	4.623	5.940
U	TACL [20]	TIP'22	18.23	0.776	0.539	2.873	4.192
U	OSMOSIS [5]	ECCV'24	17.38	0.752	0.531	2.793	4.095
U	**MUD-UIE**	–	**22.91**	**0.955**	<u>0.784</u>	**5.185**	**6.704**

generalization across diverse underwater environments.

B. Qualitative Evaluation

Figure 2 illustrates qualitative comparisons between MUD-UIE and representative competing methods on UIEB, UVEB, and LSUI, with emphasis on color restoration, visibility improvement, contrast enhancement, and structural consistency. While several supervised and semi-supervised approaches achieve visually pleasing results in specific scenarios, they often introduce color shifts, noise amplification, or spatial artifacts, particularly in regions affected by strong depth variation. In contrast, MUD-UIE produces more balanced and natural enhancements across all datasets, improving visibility while preserving stable colors and coherent structures. This behavior reflects the decision-oriented nature of the proposed approach, in which RGB appearance and depth information jointly guide the enhancement strategy and parameter prediction, rather than relying on direct end-to-end image synthesis.

C. Ablation Study

We further analyze enhancement behaviors under different dominant water appearances (green, blue, yellow, brown, and turbid) by evaluating several fixed enhancement combinations (Comb1–Comb4) and the UWE method [7], on representative subsets from UIEB, UVEB, and LSUI. As shown in Table IV, the best-performing configuration varies across conditions: green-dominated scenes often benefit from UWE-style enhancement, blue waters favor classical operator combinations that balance color correction and contrast, while yellow, brown, and highly turbid conditions typically require stronger color compensation or local contrast adjustments. These results indicate that optimal enhancement behavior is highly dependent on the underlying degradation characteristics, motivating adaptive strategy and parameter prediction, which is the core principle of MUD-UIE and explains its stable behavior across diverse underwater scenarios.

TABLE IV
ABLATION BY WATER CONDITION (BEST PER BLOCK IN BOLD).

Water	Method	P	S	UQ	QM	SM
Green	Comb1	19.2	0.81	**0.88**	3.40	3.90
	Comb2	13.1	0.66	0.52	2.85	4.25
	Comb3	16.4	0.58	0.46	2.10	3.70
	Comb4	20.3	0.79	0.71	4.35	5.95
	UWE	**23.8**	**0.92**	0.82	**5.60**	**6.80**
Blue	Comb1	20.6	0.74	0.71	3.70	4.20
	Comb2	19.4	0.69	0.55	3.10	4.55
	Comb3	17.1	0.60	0.49	2.30	3.95
	Comb4	**24.5**	**0.93**	**0.85**	**5.75**	**6.90**
	UWE	10.0	0.88	0.78	2.10	3.40
Yellow	Comb1	15.8	0.52	0.44	1.05	2.55
	Comb2	16.6	0.56	0.47	1.30	2.80
	Comb3	**22.9**	0.85	0.63	2.97	4.06
	Comb4	22.7	**0.89**	**0.76**	**5.10**	**6.20**
	UWE	20.4	0.81	0.69	4.55	5.85
Brown	Comb1	18.9	0.79	0.63	4.20	5.10
	Comb2	**22.1**	**0.91**	**0.74**	**5.30**	**6.35**
	Comb3	14.7	0.59	0.46	2.15	3.65
	Comb4	17.3	0.72	0.58	3.60	4.80
	UWE	15.5	0.68	0.53	3.05	4.10
Turbid	Comb1	**21.4**	**0.88**	**0.69**	**4.95**	**5.90**
	Comb2	20.9	0.85	0.66	4.60	5.65
	Comb3	15.2	0.62	0.48	2.40	3.80
	Comb4	18.1	0.74	0.55	3.85	4.90
	UWE	16.0	0.70	0.51	3.10	4.20

P: PSNR, S: SSIM, UQ: UCIQE, QM: UIQM, SM: UISM.

V. CONCLUSION

In this paper, we proposed MUD-UIE, an unsupervised depth-aware multimodal approach for underwater image enhancement that formulates UIE as a decision and parameter estimation problem guided by blind-reference quality assessment. By jointly exploiting RGB appearance and depth cues,

the method achieves adaptive and scene-aware enhancement without relying on paired reference images. Experimental results on UIEB, UVEB, and LSUI demonstrate that MUD-UIE delivers competitive or superior performance compared to state-of-the-art supervised, semi-supervised, and unsupervised methods while exhibiting strong generalization in diverse underwater conditions. These results highlight the effectiveness of combining depth-aware multimodal learning with blind-reference supervision for robust UIE.

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